

A Dominant Polynomial Parametrization of $xyz + t^2 + 1 = 0$

Abstract

We exhibit explicit polynomials $X, Y, Z, T \in \mathbb{Z}[a, b, c]$ satisfying $XYZ + T^2 + 1 = 0$ identically. The construction comes from an iterated Gaussian-norm factorization that preserves imaginary part 1. The resulting three-parameter polynomial map is dominant on the hypersurface, and a nonconstant one-parameter specialization is also recorded.

1 Main result

Theorem 1.1. *Let a, b, c be independent parameters, put*

$$R = a + b(a^2 + 1), \tag{1}$$

and define

$$\begin{aligned} Z &= a^2 + 1, \\ Y &= (1 + ab)^2 + b^2, \\ T &= R + c(R^2 + 1), \\ X &= -((1 + cR)^2 + c^2). \end{aligned}$$

(2)

Then $X, Y, Z, T \in \mathbb{Z}[a, b, c]$ and

$$XYZ + T^2 + 1 = 0 \tag{3}$$

identically. Hence every $(a, b, c) \in \mathbb{Z}^3$ gives an integer solution $(x, y, z, t) = (X, Y, Z, T)$ of $xyz + t^2 + 1 = 0$. For every such specialization,

$$X < 0, \quad Y > 0, \quad Z > 0.$$

Eliminating the auxiliary symbol R , the map is explicitly

$$\begin{aligned} X &= -\left(\left(1 + c(a + b(a^2 + 1))\right)^2 + c^2\right), \\ Y &= (1 + ab)^2 + b^2, \\ Z &= a^2 + 1, \\ T &= a + b(a^2 + 1) + c\left((a + b(a^2 + 1))^2 + 1\right). \end{aligned} \tag{4}$$

2 Proof

The following identity is the algebraic mechanism behind the parametrization.

Lemma 2.1 (Norm splitting). *For arbitrary elements U, V of a commutative ring, put*

$$S = U + V(U^2 + 1).$$

Then

$$S^2 + 1 = (U^2 + 1)((1 + UV)^2 + V^2). \tag{5}$$

Proof. In the universal polynomial ring with an element i satisfying $i^2 = -1$, one has

$$S + i = (U + i)(1 + UV - Vi). \quad (6)$$

The real part of the product on the right is

$$U(1 + UV) + V = U + V(U^2 + 1) = S,$$

and its imaginary part is $(1 + UV) - UV = 1$. Multiplication by the conjugate of (6) gives (5). Since this is a polynomial identity over $\mathbb{Z}$, it remains valid after specialization to any commutative ring. $\square$

Proof of Theorem 1.1. Apply Lemma 2.1 first with $(U, V) = (a, b)$. By (1) and (2),

$$R^2 + 1 = (a^2 + 1)((1 + ab)^2 + b^2) = ZY. \quad (7)$$

Apply the lemma a second time with $(U, V) = (R, c)$. Since $T = R + c(R^2 + 1)$,

$$\begin{aligned} T^2 + 1 &= (R^2 + 1)((1 + cR)^2 + c^2) \\ &= ZY((1 + cR)^2 + c^2) \\ &= -XYZ. \end{aligned}$$

This proves (3).

For integral a, b, c , one has $Z = a^2 + 1 \geq 1$. The expression $Y = (1 + ab)^2 + b^2$ cannot vanish, since $b = 0$ would leave the first square equal to 1; hence $Y \geq 1$. The same argument shows that $(1 + cR)^2 + c^2 \geq 1$, so $X \leq -1$. $\square$

3 How the construction was found

Writing the equation as

$$t^2 + 1 = -xyz \quad (8)$$

shows that one needs to split the norm $t^2 + 1 = N(t + i)$ into three polynomial factors. Multiplication by a general Gaussian integer does not preserve the special form “real part plus i ”. The key is to choose a multiplier whose imaginary cross-terms cancel.

Starting from $U + i$, multiply by $1 + UV - Vi$. Then

$$(U + i)(1 + UV - Vi) = (U + V(U^2 + 1)) + i.$$

Thus the imaginary part remains exactly 1, while taking norms gives

$$(U + V(U^2 + 1))^2 + 1 = (U^2 + 1)((1 + UV)^2 + V^2).$$

The output is again a norm of the same shape, so the operation can be iterated.

The first iteration, with $(U, V) = (a, b)$, produces

$$R^2 + 1 = ZY.$$

The second, with $(U, V) = (R, c)$, produces

$$T^2 + 1 = (R^2 + 1)((1 + cR)^2 + c^2) = ZY((1 + cR)^2 + c^2).$$

Taking X to be the negative of the final factor gives $T^2 + 1 = -XYZ$. The parametrization is therefore a two-stage composition of Gaussian-norm splittings, rather than an isolated polynomial expansion.

4 Dominance and scope

Let

$$\mathcal{H} = \{(x, y, z, t) \in \mathbb{A}_{\mathbb{Q}}^4 : xyz + t^2 + 1 = 0\}.$$

Theorem 1.1 defines a polynomial map

$$\Phi : \mathbb{A}_{\mathbb{Q}}^3 \longrightarrow \mathcal{H}, \quad (a, b, c) \longmapsto (X, Y, Z, T).$$

Proposition 4.1. *The map Φ is dominant; equivalently, its image is Zariski dense in $\mathcal{H}$.*

Proof. Consider the coordinate functions (Z, Y, T) . From (2) and (7),

$$\frac{\partial Z}{\partial a} = 2a, \quad \frac{\partial Y}{\partial b} = 2R, \quad \frac{\partial T}{\partial c} = R^2 + 1 = YZ.$$

The relevant Jacobian is therefore

$$J_{\Phi} := \det \left(\frac{\partial(Z, Y, T)}{\partial(a, b, c)} \right) = \det \begin{pmatrix} 2a & 0 & 0 \\ \partial_a Y & 2R & 0 \\ \partial_a T & \partial_b T & YZ \end{pmatrix} = 4aRYZ. \quad (9)$$

It is not the zero polynomial; at $(a, b, c) = (1, 0, 0)$ it equals 8. Hence Φ has generic rank 3, and the closure of its image has dimension 3.

The polynomial $xyz + t^2 + 1$ is irreducible over $\mathbb{Q}$. Indeed, as a polynomial in x over the unique factorization domain $\mathbb{Q}[y, z, t]$, it is primitive because yz and $t^2 + 1$ are coprime. Over the fraction field $\mathbb{Q}(y, z, t)$ it is linear and therefore irreducible; Gauss's lemma gives irreducibility over $\mathbb{Q}[y, z, t]$. Thus $\mathcal{H}$ is an irreducible hypersurface of dimension 3. The Zariski closure of $\Phi(\mathbb{A}_{\mathbb{Q}}^3)$ is irreducible and has dimension 3; because it is contained in $\mathcal{H}$, it must equal $\mathcal{H}$. This proves dominance. $\square$

Remark 4.2. Dominance says that the construction is full-dimensional, not merely a curve or surface in $\mathcal{H}$. It does not claim that every integer point of $\mathcal{H}$ is attained by integral parameters.

5 A one-parameter specialization

Taking $a = n$, $b = -1$, and $c = 1$ gives a family in which all four coordinates are nonconstant.

Corollary 5.1. *For every $n \in \mathbb{Z}$, set*

$$\boxed{\begin{aligned} x &= -((n^2 - n)^2 + 1), \\ y &= (n - 1)^2 + 1, \\ z &= n^2 + 1, \\ t &= n^4 - 2n^3 + 2n^2 - n + 1. \end{aligned}} \quad (10)$$

Then $xyz + t^2 + 1 = 0$.

Proof. Here

$$R = n - (n^2 + 1) = -n^2 + n - 1.$$

Substitution into (2) gives (10), and Theorem 1.1 applies. Equivalently,

$$\begin{aligned} & (n^4 - 2n^3 + 2n^2 - n + 1)^2 + 1 \\ &= ((n^2 - n)^2 + 1)((n - 1)^2 + 1)(n^2 + 1). \end{aligned}$$

$\square$

For example, $n = 2$ gives $(x, y, z, t) = (-5, 2, 5, 7)$, and

$$(-5)(2)(5) + 7^2 + 1 = 0.$$